

FloodWatch GH

Software Requirements Document

Community Flood Monitoring and Reporting System

Project Overview

FloodWatch GH is a web-based community flood reporting and severity prediction platform built for Ghanaian communities. This document outlines all software and hardware requirements used to develop, run, and contribute to the project.

Software Requirements

Development Tools

Tool	Version / Details	Purpose
Visual Studio Code	Latest stable	Primary code editor with TypeScript and Tailwind support
Git	2.47+	Version control for tracking all code changes
GitHub	github.com/Brefo205/Flood-Watch-GH	Remote repository and team collaboration
Node.js	v20 LTS or higher	JavaScript runtime required to run Next.js
npm	Included with Node.js	Package manager for installing project dependencies
Postman	Latest stable	Testing and debugging the backend API routes
Google Chrome	Latest stable	Primary browser for development and testing

Frontend Technologies

Technology	Version	Purpose
Next.js	16.2.10	React framework with App Router and API routes
React	19.x	UI component library
TypeScript	5.x	Typed JavaScript for code reliability and error prevention
Tailwind CSS	4.x	Utility-first CSS framework for styling
Leaflet.js	Latest	Interactive map rendering with severity pin overlays
Lucide React	0.383+	Icon library used throughout the UI

shadcn/ui	Latest	Accessible UI component system (buttons, inputs, dialogs)
react-leaflet	Latest	React wrapper for Leaflet.js map integration

Backend Technologies

Technology	Version	Purpose
Next.js API Routes	16.2.10	REST API endpoints running on Node.js
Cookie-based Auth	Custom implementation	Secure user authentication and session management
Next.js Middleware	16.2.10	Route protection and role-based access control
OpenTopography API	REST API (free)	Fetches NASA SRTM elevation data for severity prediction
D8 Flow Direction Algorithm	TypeScript implementation	Calculates flood spread zone from elevation data

Authentication and Access Control

Feature	Implementation	Details
User Registration	Custom API route	Stores name, email and password for resident accounts
User Login	Custom API route	Sets secure cookie with email, name and role
Admin Login	Hardcoded credentials	Separate admin email and password for NADMO officers
Route Protection	Next.js Middleware	Redirects unauthenticated users to login page
Role Separation	Cookie-based roles	Admin and resident roles stored in auth cookie
Logout	API route + dialog	Clears auth cookie with confirmation dialog

Database (Phase 2 — Planned)

Technology	Version	Purpose
PostgreSQL	15+	Primary relational database for persistent report storage
PostGIS	3.x	Geospatial data extension for location-based queries
Prisma ORM	Latest	Database schema management and type-safe queries

Note: The current prototype uses a structured static data layer in lib/data.ts to simulate flood reports across five Greater Accra communities. PostgreSQL with PostGIS will replace this in Phase 2.

External Services and APIs

Service	Type	Purpose
OpenTopography API	Free REST API	NASA SRTM elevation data for severity prediction
Cloudinary	Free tier	Photo upload and storage for flood reports (Phase 2)
Vercel	Free tier	Frontend hosting and deployment
Railway	Free tier	Backend and database hosting (Phase 2)

Recommended VS Code Extensions

Extension	Purpose
Tailwind CSS IntelliSense	Autocomplete and linting for Tailwind CSS classes
Prettier — Code Formatter	Automatic consistent code formatting on save
ESLint	JavaScript and TypeScript code linting
GitLens	Enhanced Git history and blame annotations
Auto Rename Tag	Automatically renames matching JSX and HTML tags
TypeScript Hero	Auto-imports and TypeScript organisation

Hardware Requirements

Minimum Requirements — Development

Component	Minimum Specification
Computer	Laptop or desktop running Windows 10/11, macOS 12+, or Ubuntu 20.04+
RAM	4GB minimum — 8GB recommended for smooth Next.js development
Storage	20GB minimum free disk space for project files and node modules
Processor	Intel Core i3 / AMD Ryzen 3 or equivalent at 2GHz+
Internet Connection	Required for OpenTopography API, GitHub pushes, and npm installs
Browser	Google Chrome, Firefox, or Edge (latest version)
Display	1280x720 minimum resolution for comfortable UI development

Recommended Requirements — Development

Component	Recommended Specification
RAM	8GB or higher for running Next.js, browser, and VS Code simultaneously
Storage	50GB SSD for faster build and compile times
Processor	Intel Core i5 / AMD Ryzen 5 or higher
Internet	Stable broadband at 5Mbps or higher
Display	1920x1080 Full HD for comfortable UI development

End User Requirements — Accessing the Website

For community members and NADMO officers using FloodWatch GH:

- Any modern smartphone, tablet, laptop, or desktop computer
- A web browser — Chrome, Firefox, Safari, or Edge (latest version)
- Internet connection — mobile data is sufficient for submitting reports
- GPS and location access enabled on device for automatic location detection
- Camera access enabled on device for uploading flood photos with reports

Admin login for testing — Email: admin@floodwatchgh.com | Password: admin123

Repository

GitHub: <https://github.com/Brefo205/FloodWatch-GH>